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# The Analysis of the Relationship Between Social Media Usage and Procrastination Among University Students

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## Introduction

Social media has become an integral part of students' daily lives, offering entertainment, communication, and access to information. However, excessive usage may negatively impact academic behaviour, particularly procrastination. This study aims to explore the relationship between social media usage and procrastination among university students and determine whether increased usage is associated with higher levels of task delay.

## Research Question

Is there a relationship between social media usage and procrastination among university students?

## Hypothesis

**“Students who spend more time on social media are more likely to procrastinate on academic tasks.”**

## Population of Interest:

Undergraduate university students in Cairo, Aging between **18-22 years old**.

## Sampling Method:

Convenience sampling through online distribution across students from different academic years.

The study used a convenience sampling method, where the survey was distributed online to university students through platforms such as messaging apps and social media. Participants were selected based on their availability and willingness to respond. To improve the representativeness of the sample, responses were collected from students across different academic years. This approach helped ensure some level of diversity within the sample while remaining practical and efficient for data collection.

## Bias Identification:

In designing this survey, several potential sources of bias were identified and considered. First, selection bias may be present since the survey was distributed online using convenience sampling, meaning that not all university students had an equal chance of participating. As shown in the data, the sample is more heavily represented by 1st and 2nd year students, which may limit how well the results reflect students in higher academic years.

Response bias may also occur, as participants may not accurately report their social media usage or procrastination behaviours, either by underestimating or overestimating them. Additionally, although efforts were made to keep all questions clear, neutral, and non-leading, slight differences in interpretation may still exist among respondents.

To reduce these biases, the survey was conducted anonymously to encourage honest responses, and the questions were carefully designed to remain simple and unbiased.

## Survey Questions:

1. What is your academic year?
2. On average, how many hours per day do you spend on social media?
3. Which social media platform do you use the most?
4. How often do you check social media while studying?
5. How often do you delay starting your academic tasks?
6. How often do you leave assignments until the last minute?
7. How would you rate your level of procrastination?
8. What do you most commonly do instead of completing a task?
9. Do you think social media affects your productivity?
10. On a scale of 1–5, how much does social media negatively impact your ability to focus?

Online survey link: <https://forms.gle/1QwE38iKfEPgsbLW6>

Number of samples collected: **110**

## Analysis:

To analyse the data collected from this survey, descriptive statistics such as the mean, median, and mode were calculated.

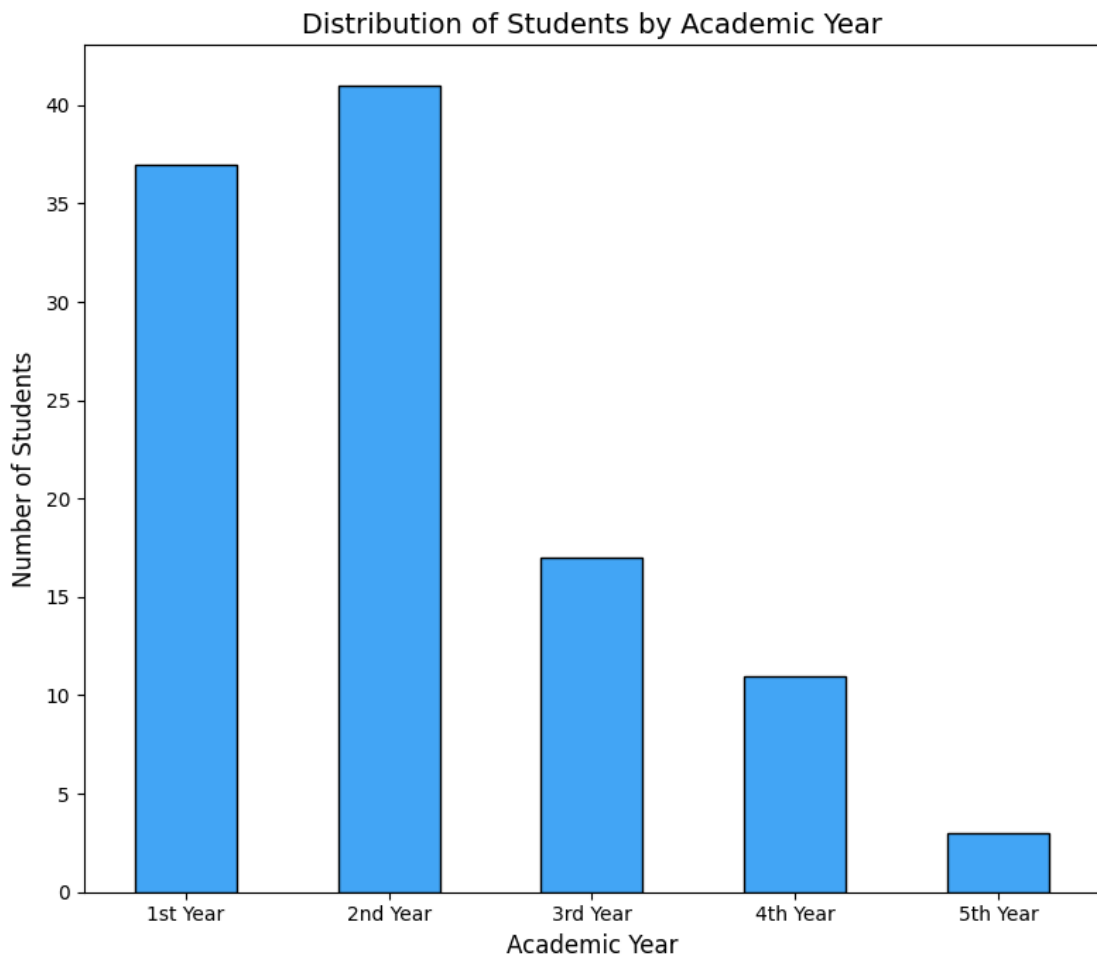
The average daily social media usage had a **mean of 4.41**, a **median of 3.50**, and a **mode of 3.5**, indicating that most students spend a moderate to high amount of time on social media.

The procrastination level had a **mean of 3.58**, a **median of 4**, and a **mode of 3**, suggesting that students generally experience moderate levels of procrastination.

Additionally, the perceived negative impact of social media on focus had a **mean of 3.98**, a **median of 4**, and a **mode of 4**, indicating that most students believe social media negatively affects their ability to concentrate.

Following this, several visualizations were created to identify patterns and relationships between social media usage and procrastination among students.

**Chart 1 — Bar Chart: Distribution of Students by Academic Year**



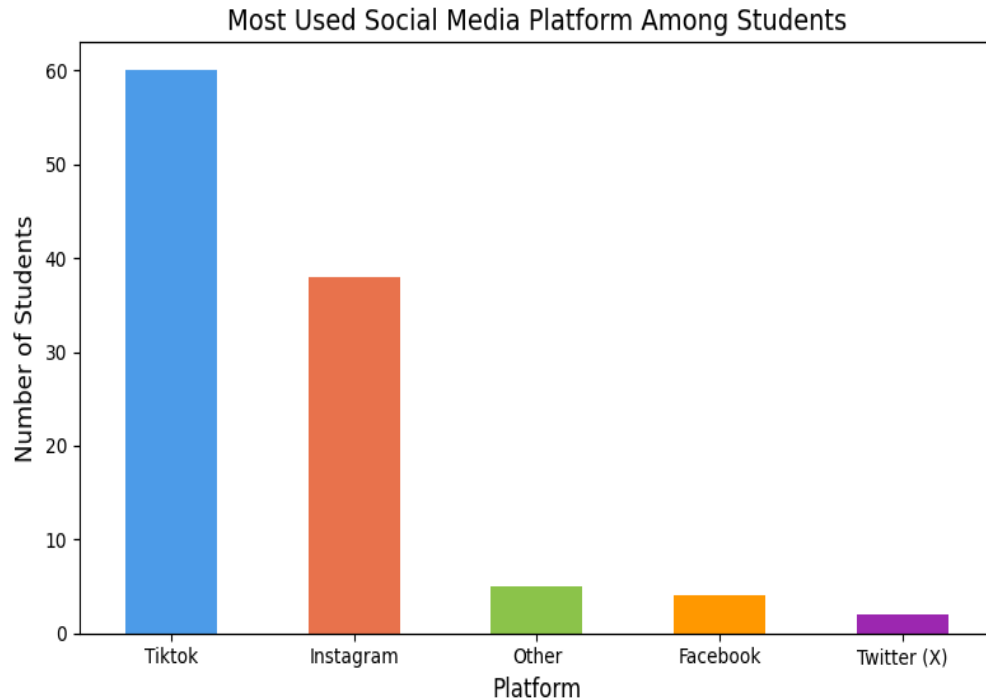
This chart shows the distribution of students across different academic years.

Most responses were collected from **2nd Year (42 students)** and **1st Year (37 students)**, while participation from higher academic years, particularly the 4th and 5th years, was noticeably lower.

This indicates that the sample is more heavily represented by early-year students, which may influence the overall findings to reflect their study habits and behaviors more strongly than those of students in advanced academic levels.

## Chart 2 — Bar Chart: Most Used Social Media Platform

Why this chart? A bar chart is best for comparing counts across categories. It shows which platforms dominate among your respondents.



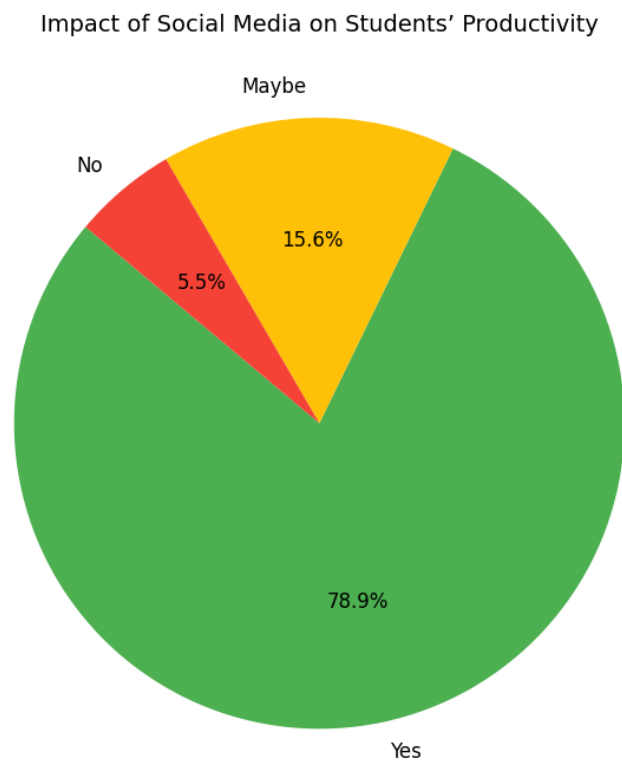
This bar chart shows the distribution of the most-used social media platforms among students.

The majority of respondents (**60 out of 109**) reported TikTok as their most-used social media platform, followed by Instagram (**38 students**), while very few students selected other platforms such as Facebook and Twitter.

This indicates that short-form content platforms dominate social media usage among university students, suggesting that highly engaging and easily accessible content may contribute more to distraction compared to other platforms.

### Chart 3 - Pie Chart: Impact of Social Media on Students' Productivity

(Does Social Media Affect Productivity?)

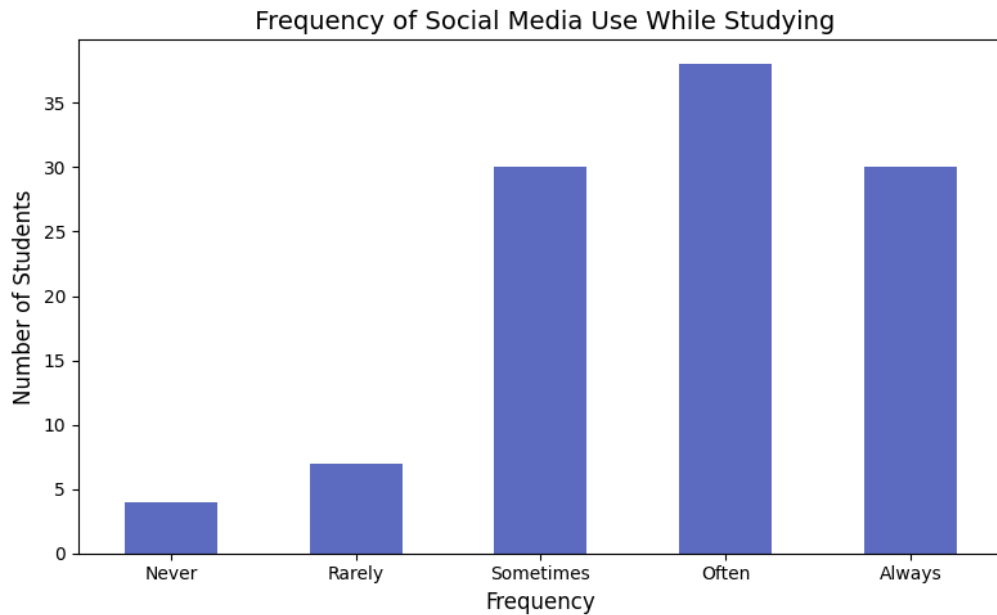


The pie chart illustrates students' opinions on whether social media affects their productivity.

An overwhelming **78.9%** of students agreed that social media affects their productivity, while only **5.5%** indicated that it does not.

This suggests a strong self-awareness among students that their social media habits negatively influence their academic performance.

**Chart 4 - Bar Chart: Frequency of Social Media Use While Studying (How Often Students Check Social Media While Studying?)**



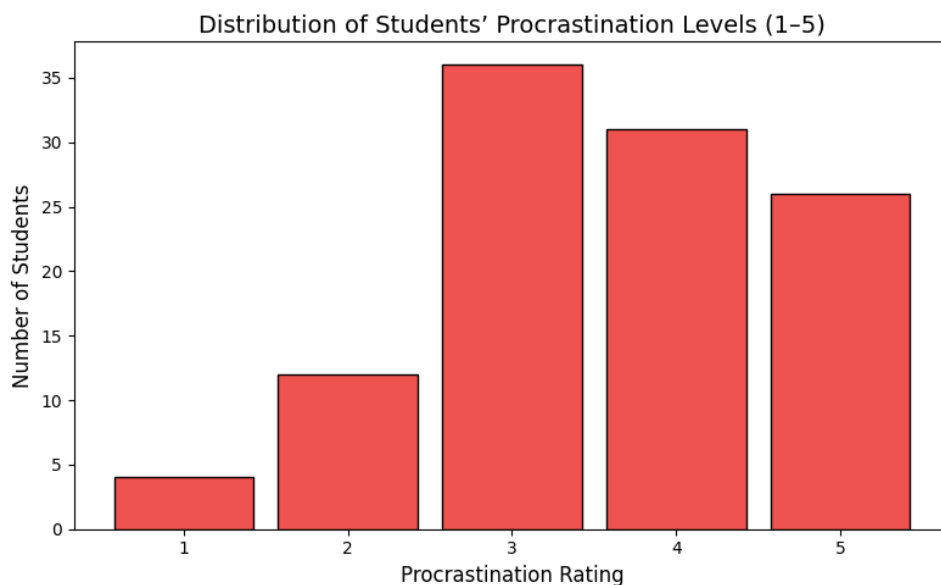
This chart shows how frequently students check social media while studying.

Most students check social media **'Often' (38)** or **'Always' (30)**, while only **4** students reported **'Never'**, indicating that social media is a frequent distraction during study sessions.

This behaviour suggests that students may struggle to maintain focus due to constant engagement with social media.

**Chart 5 — Histogram: Distribution of Students' Procrastination Levels**

Why histogram? It shows the distribution/spread of a numeric variable.

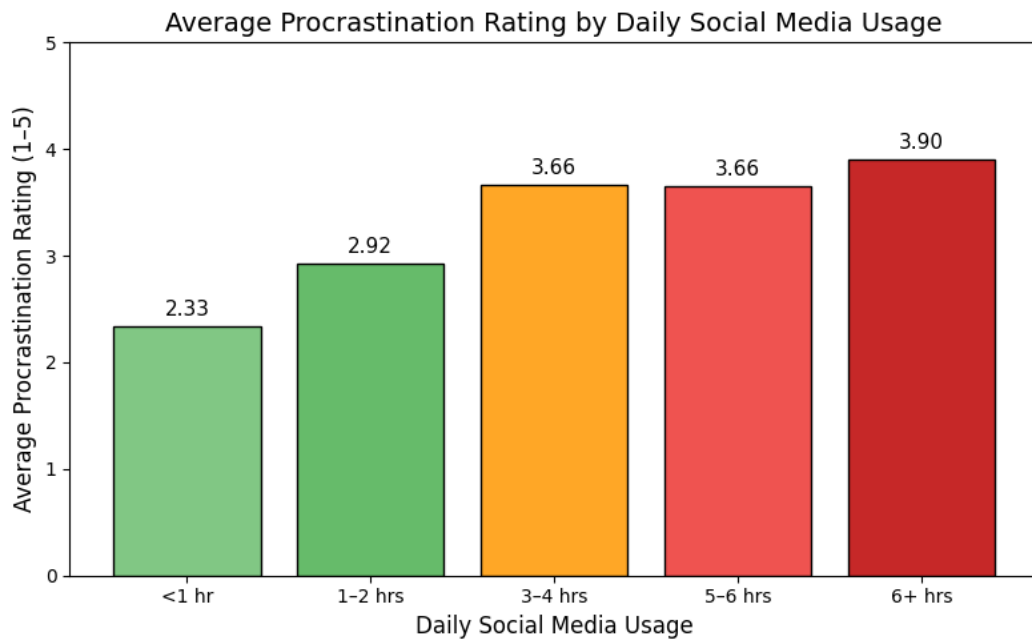


The histogram shows the distribution of procrastination levels among students.

The distribution is skewed toward higher values, with ratings of **3, 4, and 5** being the most common. The mean procrastination rating is **3.58 out of 5**, indicating that most students perceive themselves as moderate-to-high procrastinators.

This suggests that procrastination is a common issue among university students in this sample.

**Chart 6 — Grouped Bar Chart: Social Media Hours vs. Average Procrastination Rating (KEY CHART for the hypothesis!)**



This chart shows the relationship between daily social media usage and average procrastination levels.

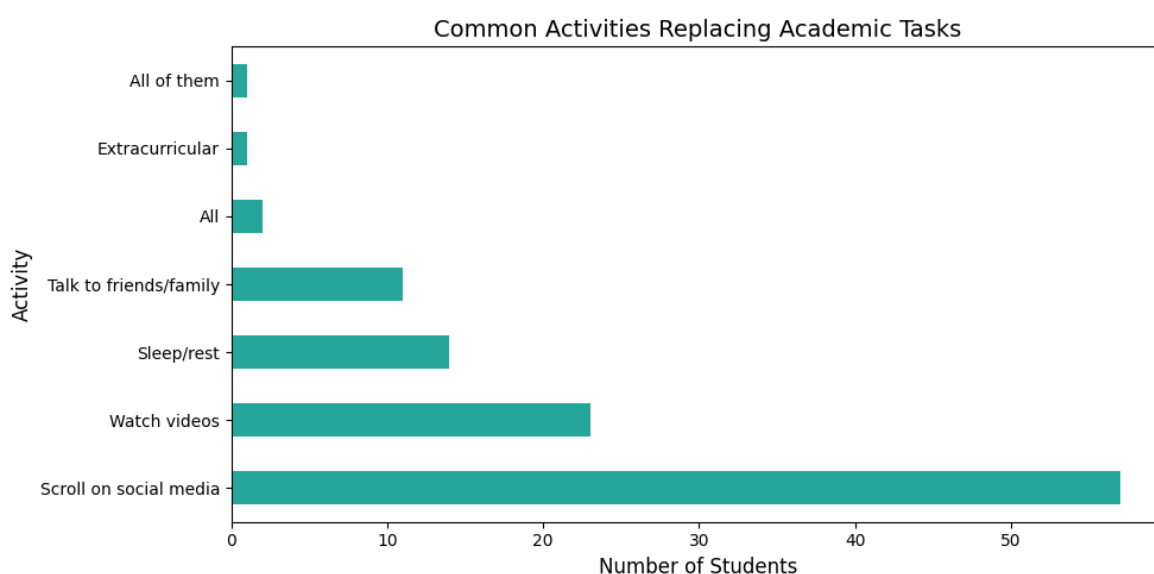
There is a general upward trend in average procrastination rating as daily social media usage increases. Students who use social media for **more than 6 hours per day** have a noticeably higher average procrastination rating compared to those who use it for **less than 1 hour**.

This pattern supports the hypothesis that higher social media usage is associated with greater procrastination among university students.



**Chart 7 — Bar Chart: Common Activities Replacing Academic Tasks**

**(What Students Do Instead of Completing Task?)**



This chart shows the activities students engage in instead of completing academic tasks.

More than half of the respondents (**57 students**) reported scrolling on social media as their primary procrastination activity, followed by watching videos (**23 students**).

This reinforces the idea that social media is a primary factor contributing to procrastination among students.

## Conclusion

The results of this study suggest that there is a relationship between social media usage and procrastination among university students. Students who reported higher daily social media usage generally showed higher levels of procrastination, as observed in the analysis.

These findings support the hypothesis that increased social media usage is associated with greater procrastination. In addition, a large proportion of students indicated that social media negatively affects their productivity, reinforcing this relationship.

However, the results are based on self-reported data and do not establish a cause-and-effect relationship. Other factors, such as time management skills or academic workload, may also influence procrastination levels. Therefore, while a clear association was observed, further research is needed to better understand this relationship.

## Any potential issues

One potential issue in this study is the use of convenience sampling, which may not fully represent all university students. The survey was distributed online, meaning that only students who chose to respond were included, which may introduce selection bias. Additionally, the data collected is based on self-reported responses, which may not always be accurate. Participants may underestimate or overestimate their social media usage or procrastination habits, leading to response bias.

Another limitation is that the survey captures general behaviours but does not account for other factors such as academic workload, stress levels, or time management skills, which may also influence procrastination. Furthermore, although most questions were designed to be neutral, there is still a possibility of slight interpretation differences among participants. Future studies could improve reliability by using a larger and more diverse sample and incorporating more objective measures of behaviour.